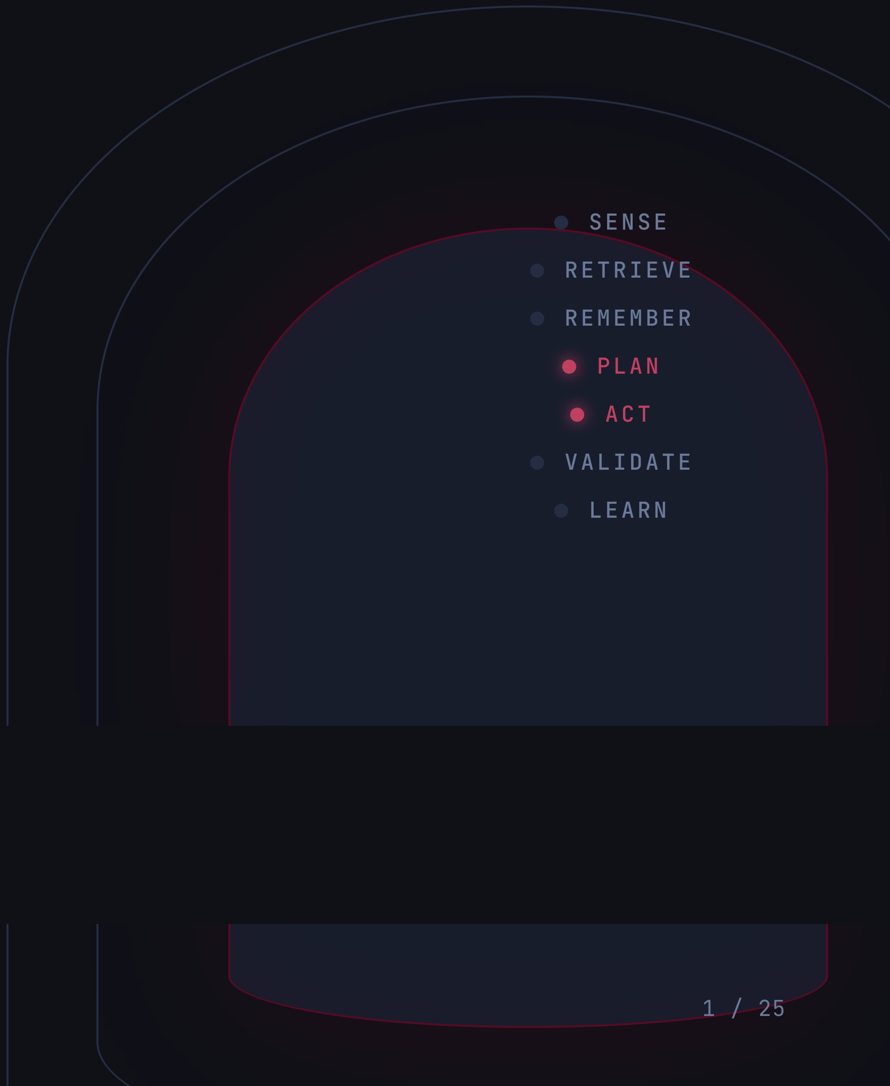
CLEW

AGENTIC DEVELOPMENT · 2026

The model is the engine. The **harness** is the car.

What makes a good agentic coding harness — and how clew keeps it coherent.



SECTION 01 / 05

The Problem

Agents can't remember between sessions — and every obvious fix for it fails.

01

THE GAP

Most harnesses only have the shallowest memory — yet memory is the moat.

Working memory
the active task — recent files, notes, distilled
compacted into the prompt each turn · lost at session end

← most stop here

Episodic memory
full transcript of this session — every request, tool call, result
saved to a file → /resume · but locked to this one thread

Long-term memory
AGENTS.md · CLAUDE.md · the artefact store
plain files · reloaded at the start of every session

the moat

2 of 3 session-bound

Working memory is distilled, then discarded at session end; episodic is a resumable transcript but locked to its own thread. Only long-term — plain files — crosses into the next session.

0 tokens / fetch

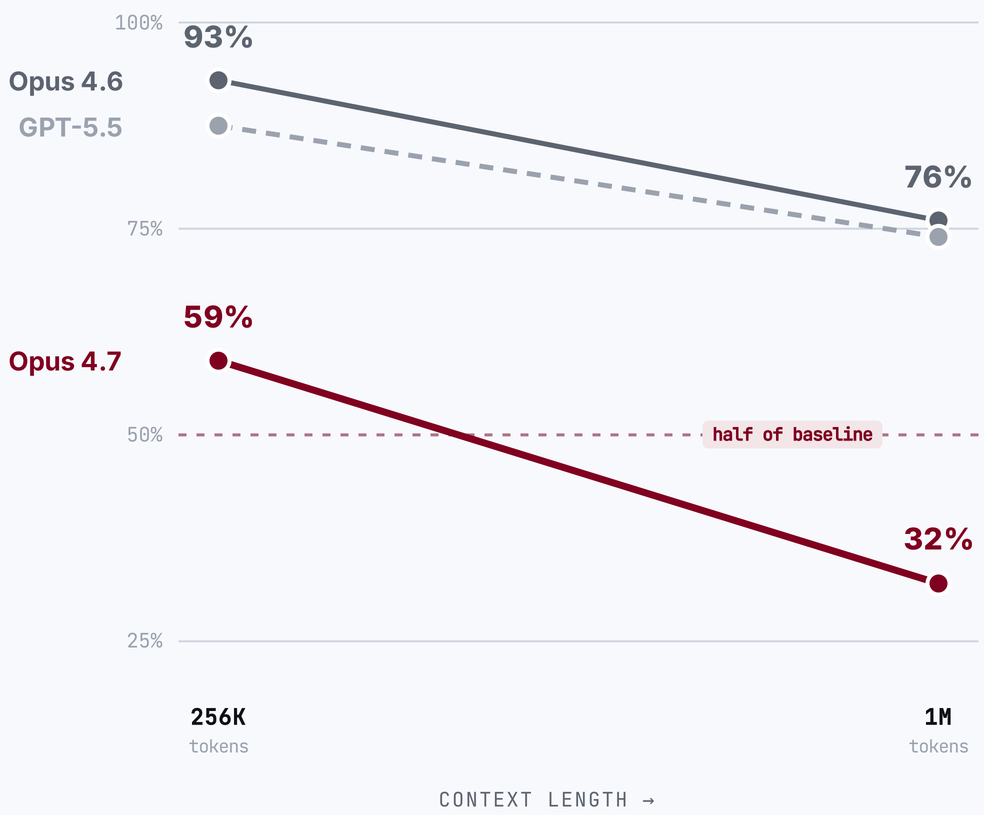
And it's a local file read — no MCP API roundtrip, no token cost, reloaded at the start of every session.

THE OBVIOUS OBJECTION *"A frontier window holds a million tokens now — so why not just let the window be the memory?"* **ANSWERED NEXT →**

No — a bigger context window is not bigger memory.

Bury one fact in a million-token prompt and ask for it back: the frontier model returns it barely **1 in 3** times. The window **forgets** as you fill it.

RETRIEVAL ACCURACY · MRCR V2



The advertised window is **not** the usable window.

59% → 32%

Claude Opus 4.7 loses ~half its multi-needle retrieval from 256K to 1M tokens. Opus 4.6 holds better (93% → 76%).

Codex too

the GPT-5.x line behind Codex CLI drops ~87% → 74% at long context — a proxy; no per-length figure is published for the Codex-tuned model.

11 of 13

models fell below half their accuracy by 32K tokens — context rot isn't new (NoLiMa, 2025).

Writing it down isn't memory either.

OpenAI shipped a 1M-line product with **zero** hand-written code — then found the obvious fix, one big context file, fails.

`1M lines of code` `0 hand-written` `1,500 PRs merged`

"One big `AGENTS.md`" failed four ways

- **Crowds out the task.** Context is scarce — the giant file pushes out the code and the docs.
- **Too much guidance becomes none.** When everything is "important," nothing is.
- **It rots instantly.** A graveyard of stale rules — the agent can't tell what's still true.
- **It can't be verified.** Drift is inevitable without mechanical checks.

If it's not in-context, it doesn't exist

Anything in Slack, Google Docs, or someone's head is invisible to the agent.

OpenAI's fix

A **~100-line table of contents** pointing into a **structured `docs/` tree** — a map, not an encyclopedia.

2025 was the "Year of the Agent." It didn't land.

The hype was model-led — but in production, the model alone fell short, expensively.

95%

of enterprise GenAI pilots showed no measurable P&L impact

MIT studied 300 deployments + 350 leaders. The blocker wasn't model IQ — it was integrating AI into real workflows.

~59%

is the best frontier model's score on realistic enterprise tasks

Scale's SWE-bench Pro — 1,865 real tasks across 41 professional repos (2026). The same models score 80–95% on the benchmark everyone quotes; the gap is the reality.

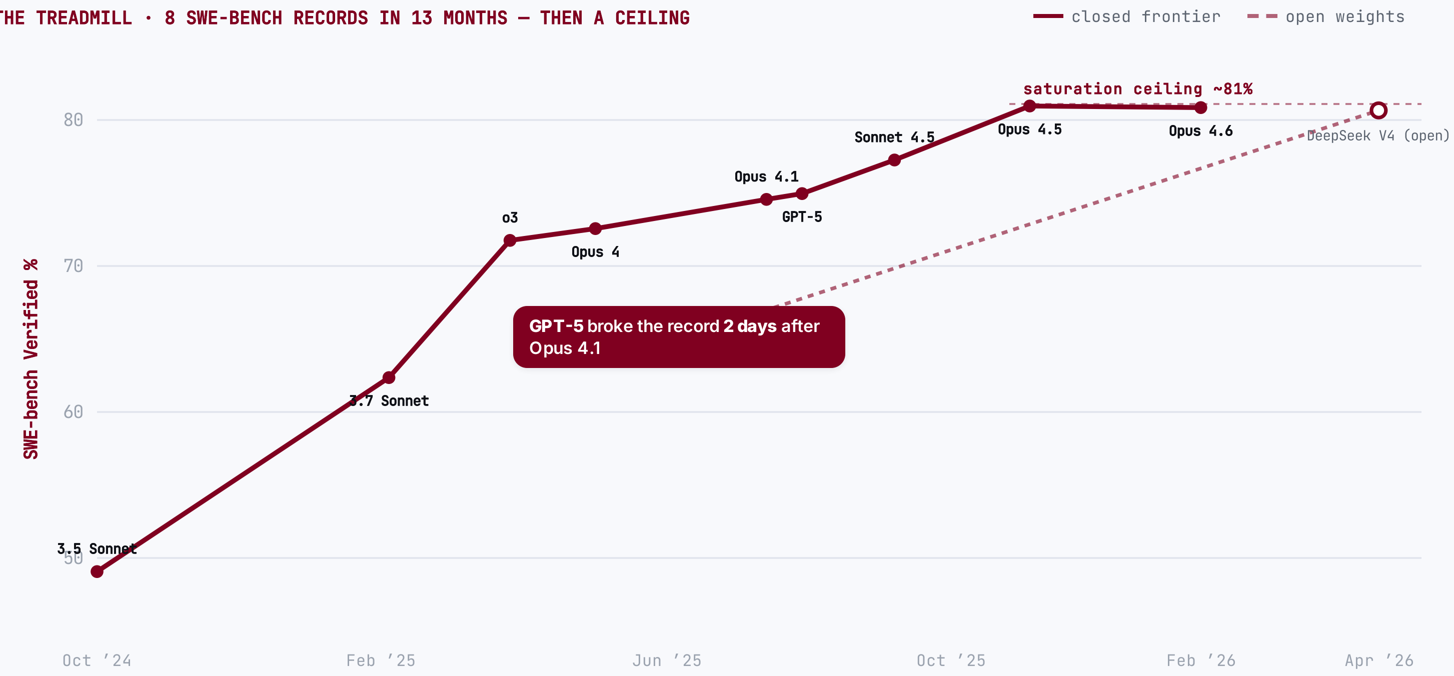
The missing variable wasn't intelligence — it was the **harness around the model**.

MIT Project NANDA · The GenAI Divide: State of AI in Business 2025 (95% of pilots — no measurable P&L impact) · via Fortune (2025) — fortune.com | Scale AI (SEAL) · SWE-bench Pro…

Models leapfrog every benchmark — then commoditize.

A new frontier model tops every benchmark... until the next does, **days** later. Then it saturates ~81%, open weights reach **parity**, and the benchmark itself gets retired.

THE TREADMILL · 8 SWE-BENCH RECORDS IN 13 MONTHS — THEN A CEILING



Open weights — now at parity

≈ 80.6% = the frontier

DeepSeek V4-Pro matches the closed SOTA on SWE-bench Verified (Apr '26). Composite gap now ~**4 months / 8 ECI pts** (Epoch AI, May '26).

The benchmark itself broke

59% of hard tests flawed

OpenAI **retired SWE-bench Verified** (Feb '26): frontier models had trained on the answers; it's saturated. → SWE-bench Pro.

Inference is collapsing

≈ 280× cheaper / 18 mo

GPT-3.5-level tokens: \$20.00 → \$0.07 per M. ~10× every year — "LLMflation."

"The models are getting commoditized."

Satya Nadella · Microsoft CEO · BG2 Pod, Dec 2024

The model is a commodity — not a moat.

Whatever tops the leaderboard this week is matched, undercut and open-weighted within months — and the benchmark itself gets retired.

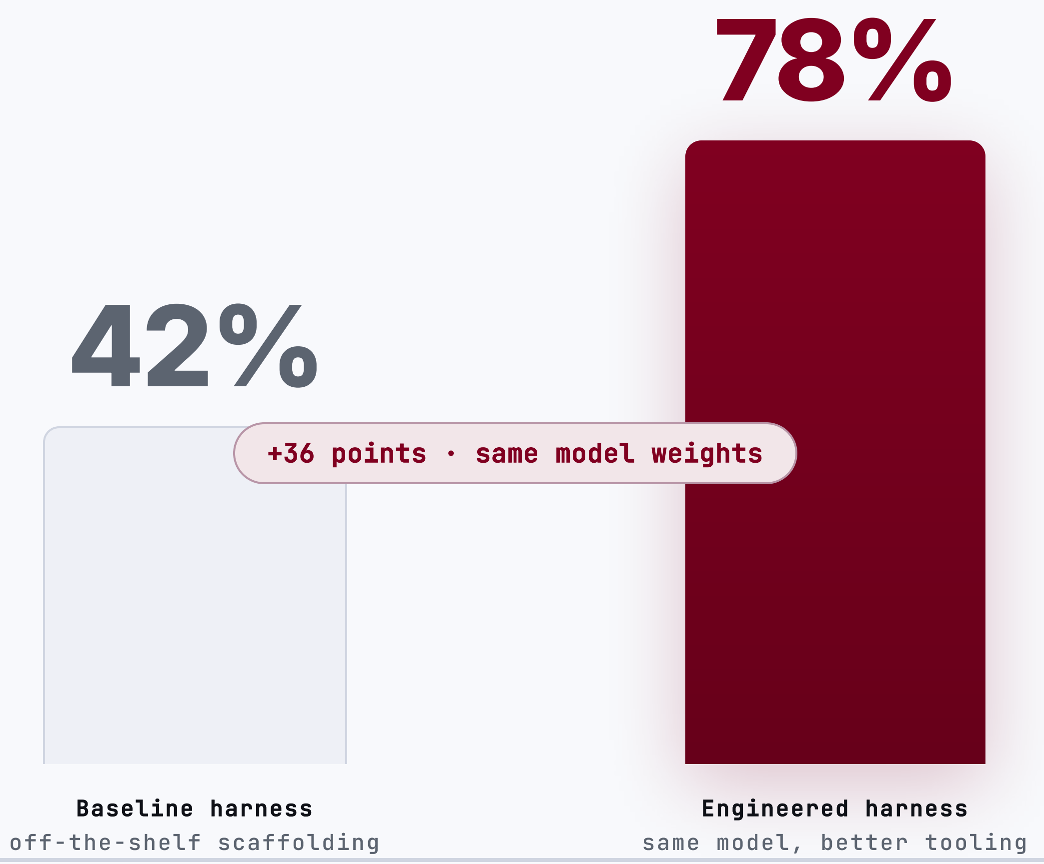
SECTION 02 / 05

The Agentic Harness

Why the tooling around the model — not the model — is the performance lever.

02

Hold the model fixed, swap the harness: 42% → 78%.

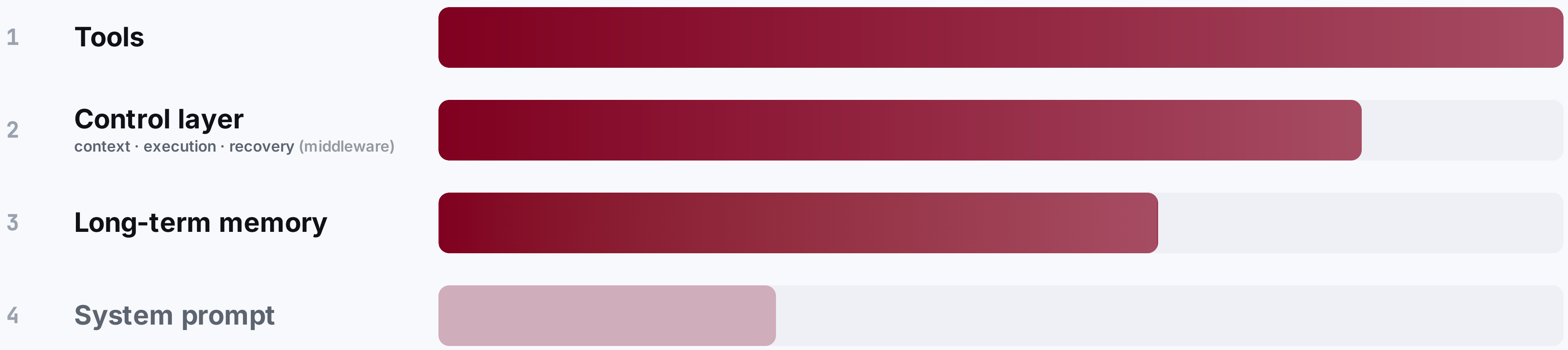


Claude Opus 4.5 · HAL CORE-Bench · the only thing that changed was the harness

THE LEVER · WHERE IT LIVES

Performance lives in the tooling around the model — not the model.

Ablation studies rank what actually moves agent performance. Model prompt-tuning sits **last**.



THE INVERSION

Most teams pour effort into #4 — the prompt. The gains sit in #1–3.

THE MODEL

Table stakes once it is capable. The harness is the differentiator.

The agent core, wrapped in *14 capabilities* — and who leads each.

② KNOW · acquire · retain · improve

Memory
Mem0 (60k) · Cognee (24k) · Letta (23k)

Knowledge
LangChain (140k) · LlamaIndex (51k) · Milvus (45k)

Learning & Adapt.
DSPy (36k) · Promptfoo (23k) · Optuna (14k)

③ DIRECT · intent & plan

Goals & Tasking
DSPy (36k) · SGLang (30k) · Guidance (22k)

Planning & Decomp.
LangChain (140k) · MetaGPT (69k) · AutoGen (59k)

① PERCEIVE · sense the world

Sensors / Perception
browser-use (101k) · Puppeteer (95k) · Playwright (92k)

Environment State
Docker (72k) · Git (62k) · Compose (38k)

Agent Core

reason · choose · loop

the model

OPUS · GPT-5 · GEMINI

④ ACT · change the world

Actuators / Actions
Daytona (72k) · Open-Interp (64k) · gh CLI (45k)

Tools & Skills
MCP servers (88k) · Composio (29k) · FastMCP (26k)

⑤ VERIFY · is the work correct?

Controls & Valid.
Playwright (92k) · Promptfoo (23k) · OpenAI Evals (19k)

Observability
LangFuse (30k) · Opik (20k) · Phoenix (10k)

⑥ GOVERN · safe & in-bounds

Safety & Govern.
Firecracker (35k) · gVisor (19k) · E2B (13k)

Identity & Access
Vault (36k) · Keycloak (35k) · Infisical (28k)

Human Feedback
Streamlit (45k) · Gradio (43k) · LangGraph (36k)

BUILDING WITH AGENTS · IN PRACTICE

Dev Tooling

The editor, model, agent-CLI and IDE you build with — open-source or proprietary — and how engineers climb from vibe-coding to real agentic engineering. The harness and the method underneath stay the same.

Open-source or proprietary — your setup, your call.

Three tiers — bring a model, compose a custom harness, or run a full IDE — each available **fully open** or **fully proprietary**. What doesn't change is the method underneath.

	OPEN-SOURCE · self-hostable	**PROPRIETARY** · hosted
Full IDE ALL-IN-ONE	VS Code ★187k · Zed ★86k · Continue ★35k · Tabby ★34k Zed standalone · VS Code + ext	Cursor · Windsurf · Antigravity · Kiro closed-source · bundled
Custom harness EDITOR + AGENT	opencode ★180k · Cline ★64k · Goose ★50k · Aider ★47k · Crush ★26k open + model-agnostic	Claude Code ★135k · Codex ★94k · Copilot · Antigravity CLI · Amp proprietary · vendor-hosted
Inference & routing OPTIONAL	Ollama ★175k · llama.cpp ★118k · vLLM ★85k · LiteLLM ★52k self-host · open	OpenRouter · Groq · Hugging Face · Bedrock · Vertex AI · Azure OpenAI managed · cloud gateway
Bring your own model THE MODEL	DeepSeek · Llama · Qwen · gpt-oss · Mistral open weights · self-hostable	Claude · GPT-5 · Gemini 3 · Grok frontier · hosted

The method
THE CONSTANT — EITHER SIDE

`spec` → `plan` → `grill-me` → `peer-review` → `loop` · artefacts · rules · skills · the **metamodel**

THE BUILD WORKFLOW · MATURITY MODEL

From “vibe coding” to *agentic engineering* — five levels.

Autonomy climbs and the harness thickens — but the human moves **up** the stack (prompter → orchestrator), never out of it.

WHAT CHANGES PER L	Ad-hoc · “vibe coding” 1	Assisted 2	Spec-driven + harness 3	Full harness · loop 4	Multi-agent fleet 5
PROCESS	Chat prompts; accept what runs	Deliberate prompts; review each diff	Spec/PRD → agent runs build/test loop	Agent owns implement→test→commit	Orchestrator → parallel agents → PRs
TOOLING	none	CLAUDE.md / AGENTS.md	docs/ · rules · skills	+ hooks · MCP · fix_plan · tests	+ git worktree per agent
HARNESS	raw model	thin: one context file	guides + first sensors (hooks)	complete + loop driver (ralph.sh)	fleet: subagents · worktrees · critic
HUMAN	**Prompter** every turn · shallow — Karpathy **vibe coding**	**Author** reviews every line — Cursor **Copilot**	**Director** approves at gates — Spec Kit · Kiro **clew**	**Supervisor** goals + escalations — **Ralph loop** Huntley	**Orchestrator** writes specs · tunes — Anthropic **+90%**

INCREASING AUTONOMY · DEEPER HARNESS →

role rises with every step ↗, never leaves the loop

⚠ **Even L5 is *supervised* autonomy, not hands-off.** METR: experienced devs were 19% *slower* on mature codebases; SWE-bench Pro tops out ~23%; Ralph’s author: “no way I’d use it on an existing codebase.” The guardrail against Thoughtworks’ **“cognitive debt”** is keeping humans on standards, architecture & final sign-off.

SECTION 03 / 05

The Metamodel

How an agent armed with skills and rules turns business context into deployable artefacts — every step traceable.

03

THE METAMODEL · THE CYCLE

Skills & rules wrap the whole lifecycle — from discovery to operations.

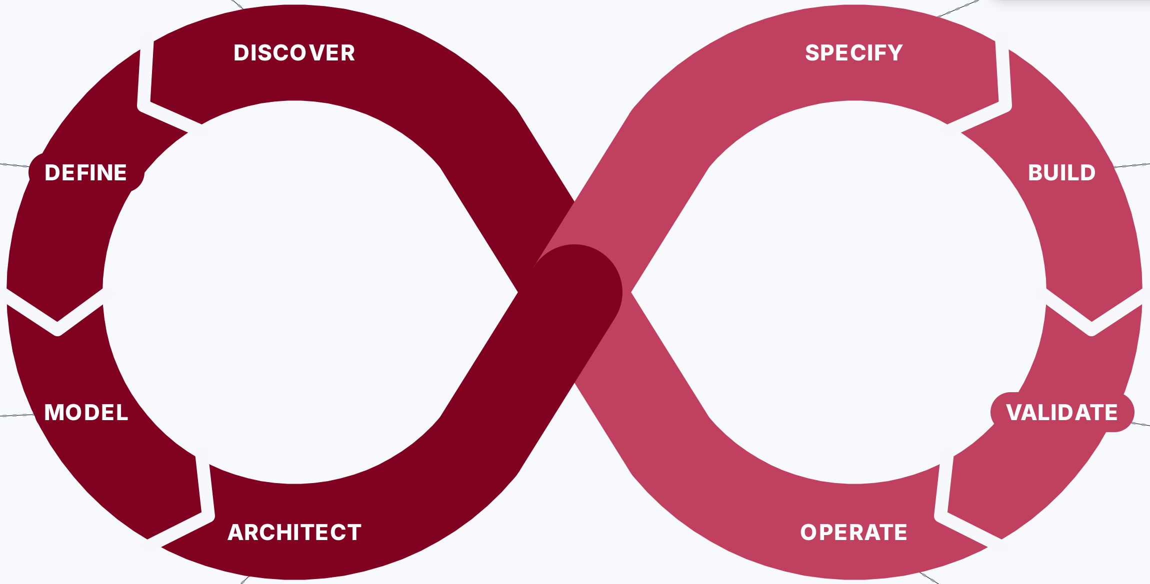
Far beyond DevOps: eight phases from business discovery to operations, each owned by a prefix, all tracing back to one metamodel.

DISCOVERY 3
Idea · Research · Workshop

SPECIFICATION 6
Functional breakdown · Use case · Quality attributes · PRD · Peer review · Grill-me

BUSINESS 9
Vision · Persona · Capability map · Value stream · Objective · Process · Model canvas · Quant model · Competitive landscape

DEVELOPMENT 8
Ralph loop · Git commit · Git worktree · PR · Git init · Agent config · Getting started · Stack guide



DOMAIN 3
Bounded context · Glossary · Domain model

QA 5 · planned
Test strategy · Test scenario · Test plan · Acceptance test · Eval harness

ARCHITECTURE 9
ADR · C4 · arc42 · UML · Structurizr · PlantUML · Service contract · CLI contract · Research

OPS 3
Runbook · Bug RCA · Terraform (Exoscale)

Discover & Define — make the context real.

WHAT THE AGENTIC ENGINEER DOES HERE

Brainstorm & capture
ideas, hypotheses

Research & scan
market, competitors

Interview & workshop
with the business

Model the business
vision → objectives

Define the domain
ubiquitous language

Business
9 SKILLS

- **Vision** — north star
- **Persona** — who we serve
- **Model canvas** — value model
- **Capability map** — what it can do
- **Competitive landscape** — market position
- **Value stream** — how value flows
- **Objective** — goals & KRs
- **Process** — operations
- **Quant model** — unit economics

Discovery
3 SKILLS

- **Idea** — capture & graduate
- **Research** — 1:1 interviews
- **Workshop** — group facilitation

Domain
3 SKILLS

- **Bounded context** — boundaries
- **Glossary** — ubiquitous language
- **Domain model** — aggregates & events

feeds → Design

Architect the solution — turn context into a solution.

WHAT THE AGENTIC ENGINEER DOES HERE

- **Decide architecture** — ADRs, C4, arc42
- **Define contracts** — service, CLI
- **Break down function** — FBS
- **Specify & qualify** — use cases, quality
- **Write & review PRDs** — one per epic

Architecture
9 SKILLS

- **ADR** — decisions
- **Research** — arch notes
- **Service contract** — API surface
- **CLI contract** — command surface
- **Structurizr** — C4 workspace
- **C4** — context → component
- **PlantUML** — UML pipeline
- **UML** — sequence/class/state
- **arc42** — arch narrative

Specification
6 SKILLS

- **Functional breakdown** — feature tree
- **Use case** — actor scenarios
- **Quality attributes** — ISO 25010
- **PRD** — one per epic
- **Peer review** — PRD/plan review
- **Grill-me** — adversarial review

UX
1 SKILL

- **Design system** — tokens & brand

Execute & Deliver — plan, build, test, ship.

WHAT THE AGENTIC ENGINEER DOES HERE

- **Sequence the roadmap**
  epics, phases
- **Plan increments**
  atomic steps
- **Build & validate**
  tests, checks
- **Deploy**
  infra, release
- **Operate & fix**
  runbooks, RCA

Specification
2 SKILLS

- **Delivery roadmap** — epics & phases
- **Implementation plan** — atomic increments

QA
5 · PLANNED

- **Test strategy** — pyramid & coverage
- **Test scenario** — from use cases
- **Test plan** — per release
- **Acceptance test** — Given–When–Then
- **Eval harness** — agent evals

Development
8 SKILLS

- **Agent config** — CLAUDE.md & rules
- **Getting started** — onboarding
- **Stack guide** — tech patterns
- **Git init** — repo setup
- **Git commit** — commits
- **Git worktree** — worktrees
- **PR** — pull requests
- **Ralph loop** — autonomous loop

Ops
3 SKILLS

- **Runbook** — operations
- **Bug RCA** — incident analysis
- **Terraform** — Exoscale infra

fed by ← Design

SECTION 04 / 05

The Build **Workflow**

From a PRD to merged code — how one engineer drives agents through alignment and verification gates, then lets them run autonomously.

04

THE BUILD WORKFLOW · END TO END

From PRD to merged code — two fresh-session gates, then autonomy.

Business context and discovery are done; the roadmap gives one PRD per epic. The loop below runs per PRD.

1 Spec ready

ALREADY DONE

- Delivery roadmap → **one PRD per epic**
- Each PRD states what must be true

PRD in hand

2 Align & plan

FRESH SESSION

- `agent-grill-me`: agent grills you ~10 Q on impl & architecture
- you answer → shared mental model
- agent writes the **implementation plan**

Plan drafted

3 Verify the plan

FRESH SESSION

- `agent-peer-review`: findings ranked critical → low
- you resolve the ones needing guidance → amend
- `ralph audit`: plan is autonomously runnable

Audit green

4 Execute & ship

SEPARATE CLI

- `agent-ralph-loop ralph.sh`: fresh agent per increment
- implement → test gate → commit, until done
- run CI on the finished work

Merged

Two **fresh-session** checks (grill-me before, peer-review after) keep me and the agent aligned; the loop only runs once the plan is proven autonomous.

PRD says what must be true; the plan says how — safely.

Two artefacts, one chain: the PRD scopes the epic, the implementation plan breaks it into runnable increments.

BUILD BY FEATURE · ONE PER EPIC

PRD

"What must be true, and why."

- **Problem, goals & non-goals** — the scope boundary
- **User stories** grounded in personas (P-NN)
- **Acceptance criteria** + success metrics
- **§0 traceability** — the epic and its dependencies

E-NN · P-NN · C-N.M.FXX · QA-XXNN · UC-NN

1 PRD
↓
1 plan

→

acceptance criteria become the plan's test gates

increments deliver the PRD scope

ATOMIC INCREMENTS · ONE PER PRD

Implementation plan

"How we get there, reversibly."

- **Ordered increments** (Inc-1 … Inc-N)
- each **small · testable · reversible**
- scope + primary files per increment
- **test gate** + exit criteria = done

Plan-NNNN · Inc-N · Status: pending → done

Two fresh agents pressure-test the work — before and after the plan.

Grill-me pulls the thinking out of my head; peer-review checks the document. Both run in a clean session.

ALIGNMENT · BEFORE THE PLAN

agent-grill-me

Live Socratic session on the PRD — one question at a time.

- Agent asks ~10 focused Q on **implementation & architecture**
- Vocabulary checked against the **domain glossary**
- Decisions crystallised into **ADRs** while context is live
- Surfaces the **thinking gaps** only questioning reveals

Output: **shared mental model** → the agent then writes the implementation plan.

VERIFICATION · AFTER THE PLAN

agent-peer-review

Static, independent scan of the finished plan.

- Reads the plan end-to-end, builds a requirement map
- Findings ranked with concrete remediation:

  critical major normal low

- I resolve the ones needing guidance → **amend the plan**

Output: **validated plan** → ready for the autonomy audit.

Peer-review finds the **mechanical** gaps automatically; grill-me surfaces the **thinking** gaps under questioning.

THE BUILD WORKFLOW · AUTONOMOUS EXECUTION

The Ralph loop — a fresh agent per increment, until the plan is done.

WHAT THE LOOP IS

- A bash script (**ralph.sh**) drives the plan to completion — one increment per iteration.
- **Fresh agent each iteration** → clean context, no drift or context rot.
- Loops while increments are **pending**; stops when every one is **Status: done**.

```
ralph.sh <workspace> \
  --max-iterations 50 --with-prd --with-push
# spawns: claude --print < iteration-prompt
```

ONE ITERATION · PERCEIVE → IMPLEMENT → VALIDATE → COMMIT

Audit must be green before the first run.

Pick next pending increment
read workspace + plan

Implement its scope
follow the primary-files list

Run the test gate
pottery wheel — up to 3 retries

Commit the increment
advance status → done

repeat with a fresh agent

All increments done → run CI

THE TAKEAWAY

Engineer the harness, not just the prompt.

Start with the highest-leverage capability: memory that doesn't rot.

`Not a PM tool` `Not a wiki` `Not a SaaS` `The repo is the source of truth`

Git for product architecture — manage what must be true, not what's in progress.